Power supplies

File: power.kicad_sch

USB-C, PD sink, XMOS, clocks

File: usb.kicad_sch

DAC, I/V, volume

File: dac.kicad_sch

Headphone amplifier and protection

File: hpamp.kicad_sch

Microphone input, phantom power, ADC

File: mic.kicad_sch

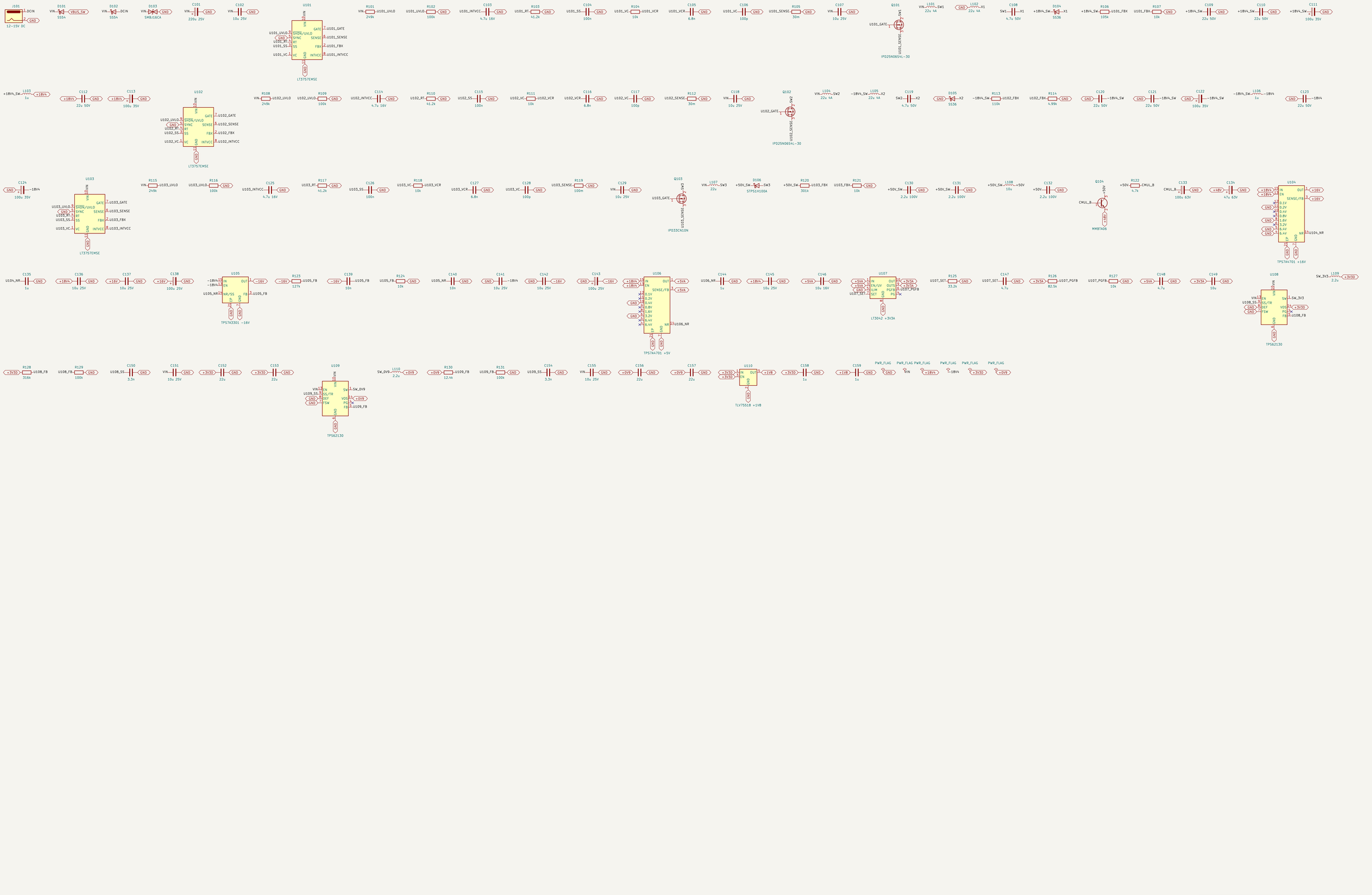
ALL-IN-ONE AUDIO INTERFACE — schematic rev A (generated by hardware/schematic/design.py)
USB-C UAC2 (XMOS XU316) | PCM1794A DAC | headphone amp >= 2 W @ 32R (OPA1612 + 2x LME49600 per ch)
XLR mic input: THAT1512 preamp 0-60 dB, switchable +48 V phantom, CS5361 ADC
Nets are joined by labels: global labels across sheets, local labels within a sheet.
See docs/design.md for the full description and the list of items to verify before layout.

All-in-one audio interface: USB DAC/amp + phantom-powered mic input
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Sheet: /
File: aio.kicad_sch

Title: All-in-one audio interface: USB DAC/amp + phantom-powered mic in

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POWER NOTES
- VIN = USB-C VBUS (after STUSB4500 load switch) OR 12–15 V DC jack, Schottky OR.
- U101 LT3757 SEPIC → +18V4, U102 LT3757 Cuk/inverting → −18V4. U103 LT3757 boost → +50V (phantom).
  LT3757: FBX = +1.6 V / −0.8 V, RT 41.2k = 300 kHz, SENSE limit 110 mV (datasheet).
- U104 TPS74A701 ANY-OUT +16.0 V (6.4+6.4+1.6+0.2 grounded). U106 TPS7A4701 +5.0 V (3.2+0.4).
- U105 TPS7A3301 −16.1 V: Vout = −1.176 V × (1 + 127k/10k). Pad and EN per datasheet (pad = GND, EN = IN).
- U107 LT3042 +3V3A clean (oscillators, converters). U108/U109 TPS62130 bucks → +3V30 / +0V9 (XMOS core).
- Q104 capacitance multiplier: +50V → ~+48.6 V phantom supply (4.7k / 100u base filter).
- VERIFY: TPS62130 FSW pin sense (GND = high freq?). MOSFET/inductor choices, LT3757 compensation.

All-in-one audio interface: USB DAC/amp + phantom-powered mic input

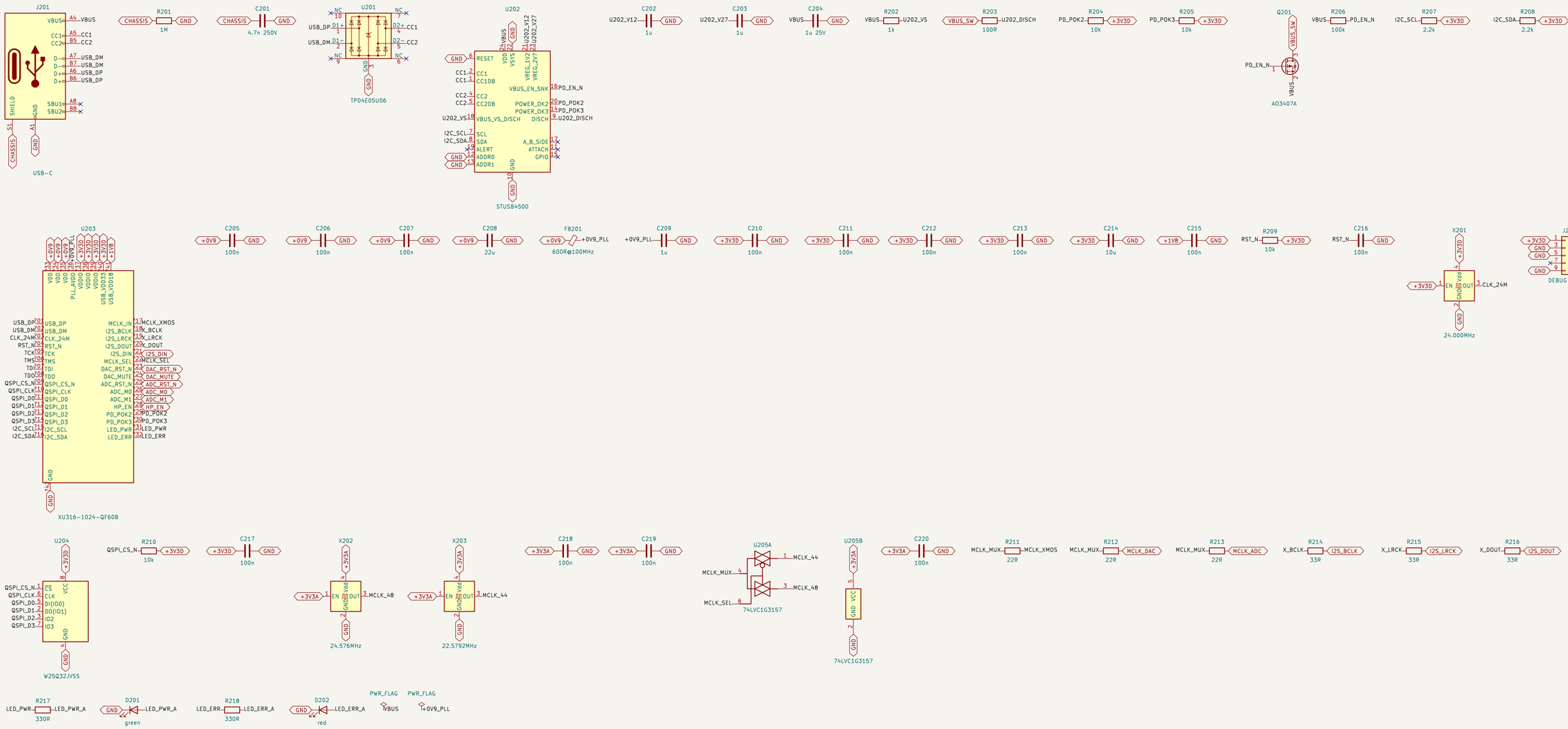
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Sheets: /Power supplies/

File: power.kicad_sch

Title: Power supplies

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USB / XMOS NOTES
- U203 XU316-1024-QF60B symbol has FUNCTION names only. Pin numbers are placeholders (?nn):
  replace with XMOS official Altium library part before PCB. Assign I2S/GPIO to ports in the .xn file.
  Known from datasheet: USB_DM=28, USB_DP=29, USB_VDD33=30, USB_VDD18=31. Core VDD = 0.9 V (0.855–0.945).
- QSPI boot flash on the XOD01/XOD04–07/XOD10 pins (check boot-mode straps in the datasheet).
- U202 STUSB4500: NVM PDO3=15V/1.5A, PDO2=12V/1.5A, PDO1=5V. VBUS_EN_SNK drives load switch Q201.
- MCLK: X202 24.576 MHz / X203 22.5792 MHz selected by U205 (S=0 -> 48k family).
- J202 debug header pinout is a placeholder: match it to your XTAG adapter.

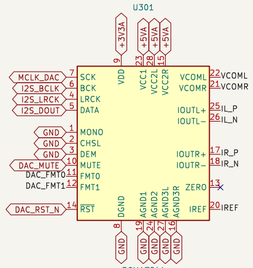
All-in-one audio interface: USB DAC/amp + phantom-powered mic input

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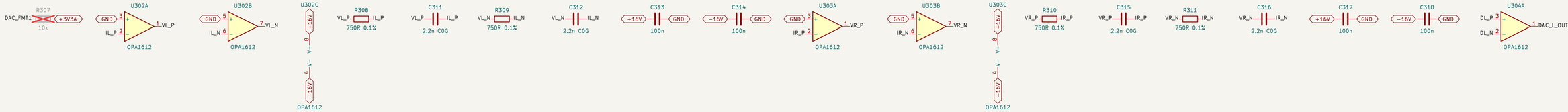
Sheet: /USB-C, PD sink, XMOS, clocks/
File: usb.kicad_sch

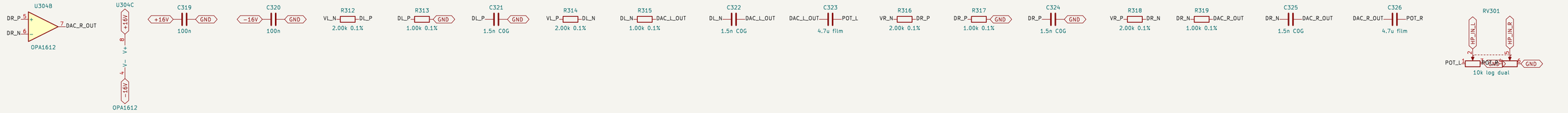
Title: USB-C, PD sink, XMOS, clocks

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DAC NOTES
- U301 PCM1794A (hardware control). Output 7.8 mA p-p per pin around −−6.2 mA.
- I/V 750R: 5.85 V p-p per leg around +4.65 V. Diff stage gain 0.5 -> -2.07 Vrms single-ended.
  Diff stage inputs are swapped (P from IOUT-, N from IOUT+) so the output is non-inverted.
- FMT0/FMT1 straps: fit pull-down OR pull-up per PCM1794A datasheet Table for 24-bit I2S (VERIFY).
- 4.7 uF film caps block DC into the volume pot. Use 0.1 % resistors in the diff stage.

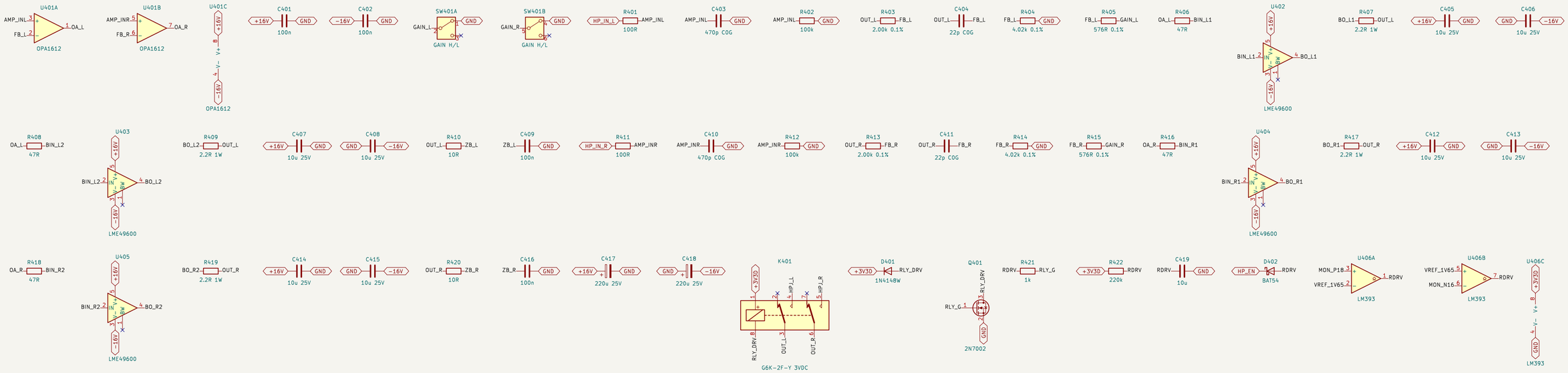
All-in-one audio interface: USB DAC/amp + phantom-powered mic input

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Sheet: /DAC, I/V, volume/
File: dac.kicad_sch

Title: DAC, I/V, volume

Size: A2	Date: 2026-10-23	**Rev: A**
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HEADPHONE AMP NOTES
- Per channel: OPA1612 + 2x LME49600 inside the loop, 2.2R ballast each, feedback from OUT_x.
- Gain: 1 + 2.00k / (4.02k || 576R) = 4.97 (SW401 closed, High) or 1.50 (open, Low).
- 2 W into 32R = 8 Vrms, 354 mA peak (177 mA per LME49600, rated 250 mA).
- Relay K401 (NO contacts) connects jacks only when RDRV is high:
  ~2–3 s after power-up (220k / 10u on RDRV), released at once if +18V4 sags (U406A), −16V collapses (U406B),
  either output has DC beyond about +−0.6 V (Q402/Q403), or XMOS pulls HP_EN low (BAT54 D402).
- LME49600 BW pin open = low-power mode (~7.3 mA Iq). Keep each TO-263 tab on >= 4 cm2 copper.

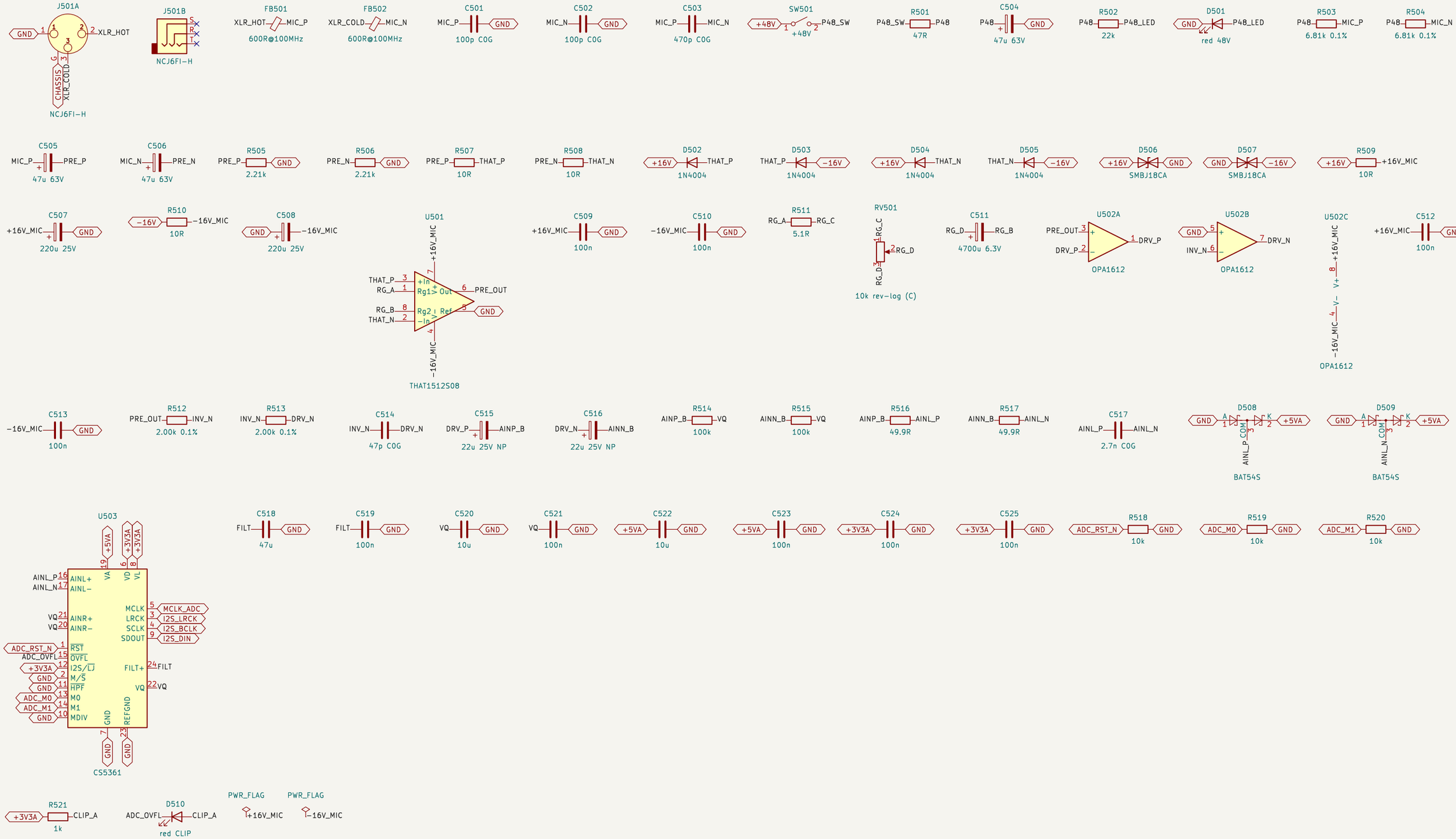
All-in-one audio interface: USB DAC/amp + phantom-powered mic input

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Sheet: /Headphone amplifier and protection/
File: hpamp.kicad_sch

Title: Headphone amplifier and protection

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MIC INPUT NOTES
- P48 phantom per IEC 61938: 6.81k 0.1 % matched feed pair, open-circuit ~48.6 V. SW501 switches it.
- Hot-plug protection: 10R series + 1N4004 clamps to the +-16V rails + SMBJ18CA TVS on the rails.
- U501 THAT1512: G = 0.5 + 5k / (5.1R + RV501) -> 0 dB to ~60 dB. 4700u cap blocks DC in the gain leg.
- U502 makes a differential signal: 1 Vrms at PRE_OUT = 2 Vrms differential = 0 dBFS on CS5361 (full scale = 1.13 x VA = 5.65 V p-p differential at VA = 5 V).
- U503 CS5361 slave mode, I2S, HPF on. M0/M1 speed mode set by XMOS per sample rate. Unused right channel tied to VQ. OVFL drives the clip LED.
- VERIFY: CS5361 FILT+/VQ capacitor values and input RC against the datasheet.

All-in-one audio interface: USB DAC/amp + phantom-powered mic input

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Sheet: /Microphone input, phantom power, ADC/
File: mic.kicad_sch

Title: Microphone input, phantom power, ADC

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